# 2022-Summer Paper 2 – Urology

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## Q1. Discuss the role of urologist in reproductive endocrinology and male infertility practice.

**Type:** LAQ (25 marks)

### Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

### 1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.
- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

### 2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–**

**300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

| Test | What it tells |
|---|---|
| LH (luteinising hormone) | Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus |
| FSH (follicle-stimulating hormone) / inhibin B | Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction |
| Oestradiol (E2) | **T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity) |
| Prolactin | Only with low T, low libido/ED (erectile dysfunction), headache or field defect |
| TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone | Only if signs; CAH (congenital adrenal hyperplasia) |

## 3. Medical (endocrine) treatment

| Condition | Mechanism | Treatment |
|---|---|---|
| **Hypogonadotropic hypogonadism** (Kallmann, pituitary) | Testis normal, signal missing: most treatable cause | **hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years |
| **Anabolic steroids / exogenous T** | Negative feedback suppresses LH → intratesticular T falls → azoospermia | Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine) |
| **Hyperprolactinaemia** | Prolactin inhibits GnRH | Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine) |
| Partial hypogonadism, obesity | Aromatisation of T to E2 suppresses LH | Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10 |
| **CAH** | Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours | Glucocorticoids |
| Primary testicular failure (Klinefelter 47,XXY) | Germ cells absent or few | No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction) |

## 4. Surgical role

| Problem | Operation |
|---|---|
| Clinical varicocele + abnormal semen | **Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele |
| Vasal or epididymal obstruction | **Vasovasostomy / vasoepididymostomy** (microsurgical intussusception) |
| EDO | **TURED** (transurethral resection of ejaculatory ducts) |
| Non-obstructive azoospermia | **Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found) |

| Problem | Operation |
|---|---|
| Unreconstructable obstruction | PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration) |
| Retrograde ejaculation; anejaculation | Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation |

## 5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

**Recent advances:** sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

**Also know:** antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

**Diagram to draw:** hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

**Key points to remember:**

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

## Q2. Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail.

**Type:** LAQ (25 marks)

*"Iscemic" is read as ischaemic nephropathy.*

### Definitions

- **RVH (renovascular hypertension):** hypertension caused by RAS (renal artery stenosis) that improves after revascularisation or nephrectomy. RAS is common; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

### Causes

| | Atherosclerotic RAS | FMD (fibromuscular dysplasia) |
|---|---|---|
| Share | Most (about 90%) | Most of the rest |
| Patient | Older, smokers, diabetic, vascular disease | Young women |
| Site | **Ostium and proximal third** (aortic plaque) | **Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type) |
| Course | Progresses to occlusion; renal function lost | Rarely occludes |
| Treatment | Medical first | **Balloon angioplasty without stent**; good BP (blood pressure) response |

**Also know:** Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, Page kidney, transplant RAS.

### Pathophysiology of RVH: Goldblatt models

- Goldblatt (1934) clipped the canine renal artery and produced hypertension. Loop: low perfusion pressure → **juxtaglomerular renin** → angiotensin II (vasoconstriction) → aldosterone (sodium retention).

| | Two-kidney, one-clip (unilateral RAS) | One-kidney, one-clip (solitary kidney or bilateral RAS) |
|---|---|---|
| Sodium | Normal contralateral kidney excretes the extra sodium (pressure natriuresis) | No normal kidney: sodium and water retained |
| Renin | **High** from stenotic side; contralateral suppressed | Rises, then returns to normal as volume expands |
| Type | **Renin (angiotensin)-dependent** | **Volume-dependent** |
| ACE (angiotensin-converting enzyme) inhibitor | Lowers BP; captopril test positive | Less BP effect; can cause **acute renal failure** |

- **Phases:** renin-dependent → sodium retention → **chronic phase**: hypertension damages the contralateral kidney (nephrosclerosis), BP no longer depends on the stenosis, and **revascularisation cures less**.
- Also: sympathetic activation, oxidative stress, endothelin, less nitric oxide.

## Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output**, far more than its metabolic need, and **autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a large translesional gradient).
- Beyond this, angiotensin II keeps GFR by efferent constriction; blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR.
- Chronic hypoperfusion → hypoxia, inflammation, microvascular rarefaction → **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive damage add injury. Early disease is reversible ("hibernating" kidney); fibrosis is not.

## Clinical clues

- Onset <30 years (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.
- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidney size difference >1.5 cm; abdominal bruit; hypokalaemia.

## Investigations

- **Duplex Doppler US:** high peak systolic velocity, renal/aortic ratio >3.5; **RI (resistive index) >0.8** predicts poor response (parenchymal disease).
- **CT angiography**; **MR angiography** (gadolinium; caution if eGFR [estimated GFR] <30 mL/min).
- **Captopril renography (DTPA, diethylenetriaminepentaacetic acid):** delayed uptake after captopril indicates renin-dependence.
- **Catheter angiography** (gold standard) with **translesional pressure gradient**; done when intervention is planned. Renal vein renin ratio >1.5 lateralises.

## Treatment

### 1. Medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** (controls renin-dependent BP; check creatinine and potassium in 1–2 weeks), calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: these patients die mainly of cardiovascular disease.

### 2. Landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting + medical therapy versus medical therapy alone; **no benefit** in renal function, BP, events or mortality; procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy; **no reduction** in cardiovascular or renal events; only a small BP fall.
- Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

### 3. Indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

### 4. Endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.

- **Ostial atherosclerotic lesion → angioplasty + stent**: balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

**5. Surgical revascularisation**

- Failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

| Technique | Use |
|---|---|
| **Aortorenal bypass** (saphenous vein, internal iliac artery in children, PTFE [polytetrafluoroethylene]) | Standard |
| **Transaortic endarterectomy** | Bilateral ostial plaque |
| **Splenorenal (left) / hepatorenal (right) bypass** | Hostile aorta |
| **Ex vivo bench repair + autotransplantation** | Branch FMD, aneurysms |
| Nephrectomy (partial or total) | Small non-functioning kidney with lateralising renin |

**Follow-up:** BP, creatinine, duplex at intervals for restenosis.

**Diagram to draw:** two-kidney one-clip versus one-kidney one-clip with the renin–angiotensin loop.

> **Key points to remember:**
> - 2K1C = renin-dependent; 1K1C = volume-dependent.
> - ASTRAL and CORAL: stenting adds nothing to medical therapy in atherosclerotic RAS.
> - FMD: balloon angioplasty without stent.
> - Revascularise for flash pulmonary oedema, declining GFR, resistant hypertension.

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## Q3a. Intravesical gene therapy.

**Type:** SAQ (10 marks)

### Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.
- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

### Agents

| Agent | Vector, mechanism | Key data |
|---|---|---|
| **Nadofaragene firadenovec** | **Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein | 75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy |
| **Cretostimogene grenadenorepvec (CG0070)** | **Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity | Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study |
| **Inodiftagene vixteplasmid (BC-819)** | Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter | Investigational |

### Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

### Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

**Also know:** non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

**Diagram to draw:** adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

**Key points to remember:**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

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# Q3b. Penile inversion vaginoplasty technique.

**Type:** SAQ (10 marks)

## Introduction

- Commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.

## Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking; continue oestrogen.

## Principle: homologous tissues

| Male structure | Becomes |
|---|---|
| Glans on dorsal neurovascular bundle | **Clitoris** |
| Preputial mucosal collar | Clitoral hood, inner labia minora |
| Dorsal bulbar urethral plate | Vestibule |
| Penile shaft skin (inverted) | **Neovaginal lining** |
| Scrotal skin (full-thickness graft) | Extra canal lining |
| Lateral scrotal skin + cord fat | Labia majora |

## Steps

```
Lithotomy; tranexamic acid 1 g; catheter
→ harvest, defat and tubularise scrotal skin graft
→ bilateral orchidectomy (leave cord fat for labial bulk)
→ deglove penis between dartos and Buck's fascia
   (keep dartos with skin = blood supply)
→ separate dorsal neurovascular bundle + glans;
   excise ALL corporal erectile tissue
→ clitoroplasty; fix at level of adductor longus tendons
→ shorten and spatulate urethra; excise bulbospongiosus
→ canal dissection: central tendon, behind prostate,
   preserve Denonvilliers fascia, repeated rectal checks
→ invert penile skin + graft into canal; labiaplasty
→ pack canal; negative-pressure dressing
```

- **Why excise all erectile tissue:** residual tissue engorges and obstructs the canal.
- Depth depends on canal dissection, not penile length. **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

## Postoperative care

- Pack and catheter out on day 5.
- **Lifelong dilation**, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy.

## Complications

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.

- **Rectovaginal fistula** (0.8% primary, 6.3% revision); urethrovaginal fistula; meatal stenosis, spraying.
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

**Diagram to draw:** sagittal view of the lined canal between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; keep Denonvilliers fascia intact.
> - Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

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# Q3c. Acute tubular necrosis.

**Type:** SAQ (10 marks)

## Definition

- ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

## Causes

| Ischaemic | Toxic |
|---|---|
| Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)** | **Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains |

## Pathophysiology

- **Why tubules:** medullary oxygen tension is only 10–20 mmHg, while the **S3 proximal segment** and **medullary thick ascending limb** consume much oxygen.
- **Loss of polarity:** $Na^+/K^+$-ATPase moves off the basolateral membrane → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole → GFR (glomerular filtration rate) falls (protective).
- Dead cells slough → **casts obstruct tubules**; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.

## Clinical course

- Initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- AKI per KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL or ×1.5, or urine <0.5 mL/kg/h for 6 hours.

## Diagnosis: ATN versus prerenal

| Index | Prerenal | ATN |
|---|---|---|
| Urine sodium | <20 mmol/L | >20 mmol/L |
| FeNa (fractional excretion of sodium) | <1% | **>1%** |
| FeUrea (on diuretics) | <35% | >35% |
| Specific gravity | >1.015 | <1.010 |
| Sediment | Bland | **Muddy-brown granular casts**, tubular cells |
| Response to fluids | Rapid | Lag of days to weeks |

- Biopsy only if recovery fails. Early biomarkers: NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1).

## Management

- Remove the cause; **avoid hypotension and nephrotoxins**; stop NSAIDs (non-steroidal anti-inflammatory drugs) and RAAS (renin–angiotensin–aldosterone system) blockers; adjust drug doses.
- Isotonic fluids; aggressive fluids in rhabdomyolysis; pre-contrast hydration.
- **Diuretics only for volume overload**; they do not treat ATN.

- Dialysis for refractory **AEIOU**: acidosis, electrolytes (hyperkalaemia), ingestions, overload, uraemia.
- Recovery takes days to weeks; mortality about half if RRT (renal replacement therapy) is needed.

**Diagram to draw:** tubular cell losing polarity → sloughing → casts and back-leak.

> **Key points to remember:**
> - FeNa >1%, specific gravity <1.010, muddy-brown casts.
> - S3 segment and thick ascending limb are most vulnerable.
> - Diuretics do not treat ATN; dialysis for AEIOU.

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## Q3d. Management of Stones in pregnancy.

**Type:** SAQ (10 marks) (lower ureteric stone in pregnancy also answered in 2020-Summer Paper 3, Q3a)

### Background

- 1 in 200 to 1 in 1500 pregnancies; **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Rising citrate, magnesium and urine flow protect, so stone rates barely change.
- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

### Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; transvaginal US for distal ureter; rise in Doppler **RI (resistive index)** suggests acute obstruction.
- Second line: **MRI (magnetic resonance imaging)** (stone as filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy**.

### Management

```
Symptomatic stone in pregnancy
 ├─ sepsis, solitary kidney, renal failure, uncontrolled pain
 │     → urgent stent or nephrostomy (+ antibiotics)
 ├─ pain controlled, no infection → observe (50–80% pass)
 └─ failed observation → ureteroscopy + holmium laser
       or temporary stent/nephrostomy until delivery
```

- **Conservative:** hydration, paracetamol, small doses of morphine. **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (third-trimester ductus closure). **MET (medical expulsive therapy)** is off-label.
- **URS (ureteroscopy):** first-line intervention for a non-complex stone in any trimester; second trimester is the safest window. Neuraxial anaesthesia, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring. Third-trimester URS may raise preterm labour.
- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents about **every 4 weeks** (4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.
- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

### After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

**Diagram to draw:** the management flowchart above.

**Key points to remember:**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks.

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## Q3e. Pre transplant nephrectomy

**Type:** SAQ (10 marks)

*Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.*

### Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

### Indications

| Group | Indication | Reason |
|---|---|---|
| **Infection** | Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis) | Sepsis under immunosuppression |
| **ADPKD (autosomal dominant polycystic kidney disease)** | No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety | Space and symptoms |
| **Malignancy** | Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash) | Most RCC (renal cell carcinoma) in recipients arises in native kidneys |
| **Hypertension** | Uncontrolled despite multiple drugs | Renin-driven |
| **Heavy proteinuria** | Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome) | Hides graft proteinuria, hypercoagulability |
| Others | Massive polyuria, large stone burden | Graft fluid balance, infection |

### Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

### Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

### Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

**Also know:** pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 $cmH_2O$.

**Diagram to draw:** enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

**Key points to remember:**

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

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## Q3f. Uretero-calycostomy.

**Type:** SAQ (10 marks) (also answered in 2020-Summer Paper 3, Q3d)

### Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

### Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

### Procedure (open, laparoscopic or robotic)

```
Flank position (45–60°); pre-placed double-J stent
→ mobilise ureter with generous periureteral tissue
→ mobilise kidney; partial lower-pole nephrectomy
   to expose the calyx widely (not simple nephrotomy)
→ ligate pelvic outlet; spatulate ureter laterally
→ apex stitch to lateral calyx wall; second at 180°
→ interrupted mucosa-to-mucosa sutures, placed then tied
→ capsule closed over cut surface, away from anastomosis
→ cover with perinephric fat, peritoneum or omentum; drain
```

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.
- Stent out at 4–6 weeks; urogram about 1 month later, then renogram. Later endoscopy needs **antegrade upper-pole access**.

### Complications

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric robotic series. The access route does not affect outcome.

**Also know:** alternatives are endopyelotomy, redo pyeloplasty, flap repair, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

**Diagram to draw:** kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

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# Rapid Revision Sheet – 2022-Summer Paper 2

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)**

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

**Q2. Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail. (LAQ)**

- 2K1C = renin-dependent; 1K1C = volume-dependent.
- ASTRAL and CORAL: stenting adds nothing to medical therapy in atherosclerotic RAS.
- FMD: balloon angioplasty without stent.
- Revascularise for flash pulmonary oedema, declining GFR, resistant hypertension.

**Q3a. Intravesical gene therapy. (SAQ)**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

**Q3b. Penile inversion vaginoplasty technique. (SAQ)**

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

**Q3c. Acute tubular necrosis. (SAQ)**

- FeNa >1%, specific gravity <1.010, muddy-brown casts.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics do not treat ATN; dialysis for AEIOU.

**Q3d. Management of Stones in pregnancy. (SAQ)**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks.

**Q3e. Pre transplant nephrectomy (SAQ)**

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

**Q3f. Uretero-calycostomy. (SAQ)**

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.